

Math 142, S14

Name:

Quiz 9

Section:

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Test the following alternating series for convergence or divergence.

$$(a) \sum_{n=1}^{+\infty} \frac{(-1)^{n-1}}{2n+1}$$

$$a_n = \frac{(-1)^{n-1}}{2n+1} \rightarrow b_n = \frac{1}{2n+1}$$

$$1) \text{ clearly } b_{n+1} = \frac{1}{2(n+1)+1} < \frac{1}{2n+1} = b_n$$

$$2) \lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1}{2n+1} = 0$$

$\Rightarrow \sum_{n=1}^{+\infty} \frac{(-1)^{n-1}}{2n+1}$ is convergent according to the alternating series Test.

$$(b) \sum_{n=1}^{+\infty} (-1)^{n-1} \frac{3n}{2n+1}$$

$$a_n = (-1)^{n-1} \frac{3n}{2n+1} \rightarrow b_n = \frac{3n}{2n+1}$$

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{3n}{2n+1} = \lim_{n \rightarrow \infty} \frac{3}{2 + \frac{1}{n}} = \frac{3}{2} \neq 0$$

Thus $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (-1)^{n-1} b_n$ does not exist

$\Rightarrow \sum_{n=1}^{+\infty} (-1)^{n-1} \frac{3n}{2n+1}$ diverges by the Divergence Test.

Problem 2. (4 points) Use the Ratio Test to determine convergence or divergence of

$$\sum_{n=1}^{+\infty} \frac{n}{3^n}$$

$$a_n = \frac{n}{3^n}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{\frac{n+1}{3^{n+1}}}{\frac{n}{3^n}} = \lim_{n \rightarrow \infty} \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right) \cdot \frac{1}{3}$$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right) \cdot \frac{1}{3}$$

$$= 1 \cdot \frac{1}{3} = \frac{1}{3}$$

$\Rightarrow \sum_{n=1}^{+\infty} \frac{n}{3^n}$ is (absolutely) convergent.